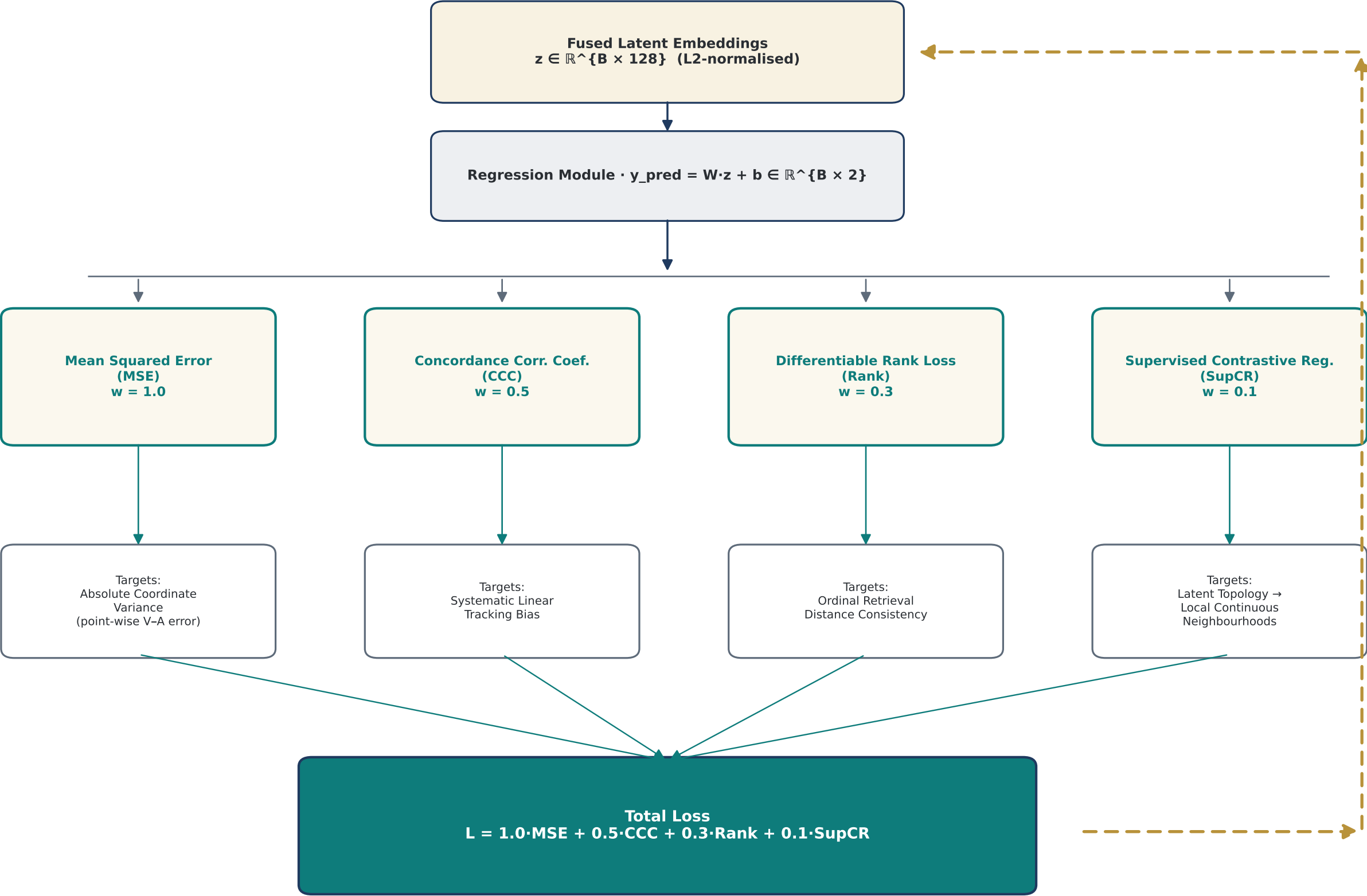


Figure 2 — Multi-Objective Hybrid Loss Architecture

Forward pass: 4 weighted loss components target distinct dataset challenges. Backward pass: differential learning rate over fusion vs head.



Gradient Backpropagation
Differential LR:
 $\eta_{\text{fusion}} = 1 \times 10^{-2}$
 $\eta_{\text{head}} = 1 \times 10^{-4}$
 $wd = 1 \times 10^{-3}$

Forward (solid) optimises 4 complementary criteria; backward (dashed gold) injects per-parameter-group learning rates.